

TOURNAMENT OF TOWNS

31st Tournament • Academic Year 2009–2010

Fall (Autumn) Round • 18 October 2009 • Senior (Grades 10–11) • Basic (O-Level)

Score is based on the three problems with the best results.

Problem 1. [4 points]

A seven-digit code consisting of seven different digits is called good. The safe's password is a good code. It is known that the safe will open if a good code is entered and at some place the digit of the code matches the corresponding digit of the password. Can the safe be guaranteed to open faster than in 7 attempts? [Diagram]

(D.V. Baranov)

Problem 2. [4 points]

In space, a closed six-edged broken line ABCDEF is located, the opposite edges of which are parallel ($AB \parallel DE$, $BC \parallel EF$, and $CD \parallel FA$). At the same time, AB is not equal to DE . Prove that all edges of the broken line lie in the same plane. [Diagram]

(V.V. Proizvolov)

Problem 3. [4 points]

Do there exist such natural numbers a, b, c, d that $a^3 + b^3 + c^3 + d^3 = 100^{100}$?

(M.V. Murashkin)

Problem 4. [4 points]

On the sides of a regular 2009-gon, a point was marked on each side. These points are the vertices of a 2009-gon with an area of S . Each of the marked points was reflected with respect to the center of the side on which this point lies. Prove that the 2009-gon with vertices at the reflected points also has an area of S . [Diagram]

(P.A. Kozhevnikov)

Problem 5. [5 points]

In a country, there are two capitals and several cities, some of which are connected by roads. Among the roads, there are toll roads. It is known that on any path from the southern capital to the northern capital, there are at least ten toll roads. Prove that all toll roads can be distributed among ten companies so that on any path from the southern capital to the northern capital, there are roads of each company.

(I.V. Netay, D.V. Baranov)